

Farah Tawalbeh

+971 543118304 | ftawalbeh1@gmail.com

[LinkedIn](#) | Nationality: Canada | Golden Visa Holder

EDUCATION

American University of Sharjah, UAE

Bachelor of Science in Computer Engineering with a minor in Data Science

August 2023 - June 2027

- **Dean's List Recipient** for **multiple** semesters

[GPA 3.55/4.00]

- Inducted into the Alpha Lambda Delta Honor Society

Relevant Coursework: *Object Oriented Programming, Computer Networks, Electronics, Data Structures & Algorithms, Operating Systems, Microcontrollers*

EXPERIENCE

American University of Sharjah

Sharjah, United Arab Emirates

Undergraduate Teaching Assistant

Aug 2025 – Jan 2026

- **Data Structures & Algorithms**

- Evaluated **150+ quizzes and assignments** for students in **Data Structures & Algorithms**, ensuring correct application of core concepts.
- **Communicated clearly** with students and faculty to support course delivery, and maintained accurate **records and documentation** for assessments.
- Evaluated implementation-oriented problems involving **recursion, graph traversal, hashing techniques, and abstract data types**.

SOC (Society of Cybersecurity)

President

Dec 2025 – Present

- **Coordinated and promoted cybersecurity events**, increasing turnout for **CTFs and technical bootcamps**.
- Simplified CTF concepts, such as **cryptography, privilege escalation, and network reconnaissance**.
- Gained **hands-on exposure to cybersecurity tools** and workflows while assisting in presenting **Linux, Python CLI tools, socket programming, PHP, and SQL** topics for workshops.
- Organized **industry visits**, collaborating with **professionals from external organizations**.

PROJECTS

Simulation of Mano's Basic Computer

- Built a full CPU simulator with **cycle-accurate execution** and full datapath modeling.
- Implemented **all Mano instruction types** with correct micro-operations and SC sequencing.
- Added **CLI and GUI interfaces** for stepping, visualization, and debugging.
- Integrated **profiler for cycles, CPI, and memory access stats**.
- Designed a **two-pass assembler for converting assembly to hex**.

Multithreaded Producer-Consumer System (POSIX Threads)

- Implemented a **multithreaded Producer-Consumer** system in C using **POSIX threads, mutexes, and semaphores** to synchronize access to a **bounded circular buffer**.
- Designed **blocking and termination logic to prevent race conditions** using a **poison-pill mechanism**.
- Measured system performance by **recording per-item latency** and overall throughput using monotonic clocks.

ARSL Dynamic Interpreter

- Developing an interactive **Arabic Sign Language Recognition (ARSL)** system for **continuous sign interpretation** using **Python**.
- Applying **machine learning and computer vision techniques** for continuous sign language recognition.
- Designing a **data preprocessing and feature extraction pipeline** for temporal hand and body movements.

ACHIEVEMENTS & EXTRACURRICULARS

- Awarded **Pearson's Student Excellence Award** for A Level and IGCSE results.
- Received '**Product of the Year**' award in the **INJAZ 2025 National Company Program Competition**.
- Participated in a **youth mentorship program** based in Ghana, where **educational workshops** were organized

SKILLS

Programming Languages: Assembly, C, C++, Python, JavaScript, HTML, CSS, MATLAB

Engineering Software: PSpice, Logisim, ELVISmx, MPLab, Cisco Packet Tracer, Linux/Unix Environments

Multimedia Software: Adobe After Effects, Adobe Premiere Pro

Office Tools: Microsoft Word, Excel, PowerPoint

LINKS

LinkedIn: <http://www.linkedin.com/in/farah-tawalbeh>

Github: <https://github.com/farahtaw07>